Machine Learning-Driven Prediction and Analysis of Mental Health Risks Among University Students

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***Abstract*—University students increasingly face psychological challenges driven by academic, social, and lifestyle pressures—issues that have intensified in the post-pandemic landscape. This study proposes a machine learning-based approach to early identification of mental health risks among undergraduates, utilising non-clinical behavioural data collected through structured surveys at a Thai university. The dataset, comprising 266 entries, was enhanced using SMOTE to balance class distribution and improve model robustness. Three classification algorithms—Random Forest, XGBoost, and Deep Learning—were trained and assessed using accuracy, precision, recall, and F1-score. XGBoost delivered the highest performance, achieving an F1-score of 83.90% and accuracy of 90.60%. Beyond classification, correlation analysis identified key behavioural and psychological predictors of mental health, while K-means clustering uncovered distinct subgroups among at-risk students. The findings underscore the potential of interpretable, data-driven methods to support proactive mental health strategies in educational environments.**

Keywords—machine learning, mental health problem, classification, correlation matrix, clustering

I. Introduction

Mental health conditions among university students around the world will have worsened due to increased academic expectations, financial instability, and the long-term psychological impacts of the COVID-19 epidemic [1]. According to recent research, approximately 70% of students report having major academic challenges, making mental health issues one of the main causes for college dropout. The severity of the crisis is highlighted by the fact that reported mental health issues among students in the UK have increased by more than 200% since 2015 [2].

Thailand's university students also struggle with severe mental health issues. According to a survey of 601 undergraduates, academic stress was a major cause of burnout, even though resilience and self-leadership reduced its effects [3]. A nationwide study of 14,621 students found that 31.4% had depressed symptoms, with poor sleep, strained relationships, and scholastic challenges being major contributors [4]. Despite growing awareness, there are still impediments that prevent students from receiving timely support in institutions, which has an impact on their academic performance, long-term prospects, and general well-being [5].

Machine learning (ML) offers a potential way to detect and intervene in student mental health issues. Prior research has shown that ML models, such as Random Forest and XGBoost, can successfully capture complex behavioral and psychological patterns, resulting in high prediction accuracy in identifying mental health problems among students [6].

Building on this, the current work applies and analyzes three supervised learning algorithms—Random Forest, XGBoost, and Deep Learning—to non-clinical behavioral survey data collected from undergraduates in Chiang Rai, Thailand. Along with classification, unsupervised clustering is used to identify behavioral subgroups among at-risk students, and correlation analysis is used to determine the most significant risk factors. In addition to improving predictive modeling, this approach provides information on student behavior that can help with focused interventions. The study is divided into five sections: methodology, results, conclusion, introduction, and connected theory and works.

This paper is organised into five main sections: Introduction, Related Theories and Works, Methodology, Results and Discussion, and Conclusion.

II. Related Theory and Works

In this section, the related theory and works applied to this study are divided into 7 parts: mental health, correlation matrix, K-means algorithm, SMOTE upsampling algorithm, random forest algorithm, XGBoost algorithm, and deep learning algorithm.

A. Mental Health

Mental health is crucial to an individual's total well-being, affecting thoughts, emotions, behaviours, and interpersonal connections. It assists individuals in coping with daily stress, making decisions, and engaging in their communities. This significance encompasses all life phases, especially during university years—a time characterised by substantial academic, emotional, and social transitions. As students transition into this phase, they frequently encounter problems such as heightened academic expectations, increasing autonomy, and detachment from their familiar support networks. Such pressures may result in elevated levels of stress, anxiety, and depression. Research constantly underscores the significance of mental well-being for student achievement and personal growth [7],[8]. A research at Taylor's University similarly identified academic overload, time limits, and future-related pressure as significant drivers to students' mental stress [9]. Consequently, comprehending and tackling these challenges is crucial for enhancing kids' academic achievement and ensuring their long-term psychological well-being. Numerous previous studies have demonstrated that machine learning is an exceptionally excellent method for forecasting the mental health of university students [10], [11]. By integrating with university support systems to evaluate extensive, intricate datasets and discern significant trends, machine learning could facilitate early identification, diagnosis, and tailored treatment techniques for at-risk patients.

B. Correlation Matrix

A correlation matrix is a statistical method used to measure the strength and direction of linear relationships between multiple variables simultaneously. It has been widely used in previous research as a starting point for exploring factors affecting students' mental health. By applying this tool, researchers can assess how variables like academic performance, sleep patterns, financial pressure, and social support relate to one another. In multivariate analysis, the correlation matrix is essential because it summarises the pairwise relationships among variables, helping to identify potential patterns and associations [12],[13]. For instance, research by Janicijevic et al. explored the link between oral and mental health among university students in Serbia, finding a strong association between poor oral health and higher levels of depression and anxiety. The study highlighted a bidirectional relationship, indicating that mental and oral health can influence each other. These results emphasise the value of correlation matrices in identifying key interconnections between variables, offering a vital basis for building reliable predictive models in mental health studies [14].

The Pearson correlation coefficient is defined as:

$$\gamma_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \overline{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2} \cdot \sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

where x_i and y_i are paired values of two variables, and $\bar{x}$, $\bar{y}$ are their means. A value close to +1 indicates strong positive correlation, while –1 indicates strong negative correlation.

C. K-Means Algorithm

K-means clustering is an unsupervised learning method that groups unlabelled data based on shared characteristics. In mental health research, it has been widely used to identify behavioural patterns or subgroups among students based on psychological traits. The algorithm works by minimising differences within each group, helping researchers uncover meaningful patterns in mental health data. For example, Liu et al. applied K-means clustering to analyse data from 1,000 college students, successfully identifying three distinct groups and enabling more targeted support strategies. This demonstrates the method's value in creating tailored, data-informed mental health interventions. [15].

The algorithm minimises the within-cluster variance:

$$J = \sum_{i=1}^{k} \sum_{x_j \in C_i} \left\| x_j - \mu_i \right\|^2$$

where C_i is the set of points in cluster i and μ_i is the centroid. This ensures that data points are grouped around the nearest cluster mean.

D. SMOTE Upsampling Algorithm

A common issue in real-world data is class imbalance, where one category greatly outnumbers others. This often leads to models that favour the majority class and fail to accurately identify important but less frequent cases. In health-related studies, such as detecting students at risk of mental health issues, this can be particularly problematic. To combat this, the SMOTE (Synthetic Minority Oversampling Technique) method is frequently used. Unlike random oversampling, which simply replicates existing data and may cause overfitting, SMOTE generates new, diverse synthetic examples by interpolating between existing minority class instances. This helps models learn more balanced and accurate decision patterns [16],[17],[18],[19]. Matharaarachchi et al. claim that SMOTE is particularly useful in medical settings since it increases classification accuracy without exaggerating the significance of noisy or unusual cases [17]. Recent research has shown that SMOTE can significantly enhance prediction accuracy in student mental health datasets. In this study, although the final dataset had a relatively balanced class distribution (147 "at-risk" and 119 "not at risk"), the overall sample size of 266 was too small for effective machine learning training. To address this, SMOTE was used to expand the dataset to 500 instances by generating synthetic samples. This step was integrated into the preprocessing phase to help improve model learning and generalisation before classifier training. [18],[19]. New samples are created using interpolation: $x_{new} = x_i + \lambda \cdot (x_{nn} - x_i)$, $\lambda \sim U(0,1)$ where x_i is a minority class instance, x_{nn} is one of its nearest neighbors, and λ is a random number between 0 and 1. This helps balance the dataset for training.

E. Random Forest Algorithm

Random Forest is an ensemble learning method that builds multiple decision trees using random feature selection and bagging to enhance accuracy and reduce overfitting. By aggregating the outputs of these trees, it generates more reliable predictions. This approach is particularly useful in mental health research due to its ability to manage complex, high-dimensional, and noisy data. It can also identify key variables influencing outcomes, making it well-suited for analysing diverse psychological and behavioural data from university students [20],[21],[22]. For instance, it has been applied to predict student stress and depression using data such as academic performance, daily routines, and behavioural patterns. It has also shown promising results in detecting students at risk of suicide and monitoring stress through mobile device usage. In this study, the algorithm was implemented using RapidMiner to classify students' mental health status based on survey responses and to pinpoint the key factors contributing to their condition [23].

The final prediction is obtained by majority voting:

$$\hat{y} = mode\{h_1(x), h_2(x), \ldots, h_T(x)\}$$

where $h_t(x)$ is the prediction of tree t, and T is the number of trees.

F. XGBoost Algorithm

Extreme Gradient Boosting (XGBoost) is an advanced and efficient version of gradient-boosted decision trees, designed for high performance and scalability. It enhances model accuracy through regularisation, parallel processing, and effective tree pruning. XGBoost is especially useful in mental health prediction due to its strong performance with imbalanced datasets—a common challenge in this field. Additionally, it supports explainable AI tools like SHAP values, allowing researchers to interpret how the model makes its predictions, which is crucial in sensitive areas such as psychological evaluation [24],[25],[25]. XGBoost has been successfully used in mental health studies to predict students' depression and anxiety levels by analysing academic performance, sleep habits, and behavioural patterns. It has shown higher accuracy compared to traditional classifiers and offers meaningful insights into key predictive features, supporting early intervention efforts. In this study, XGBoost was applied to enhance precision and recall in identifying students' mental health conditions by learning from both behavioural and academic data [26],[27]. The objective function combines the training loss with a regularization term:

$$\mathcal{L}(\phi) = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k)$$

where $l(y_i, \hat{y}_i)$ is the loss function (e.g., logistic loss) and $\Omega(f_k)$ controls model complexity.

G. Deep Learning Algorithm

Deep learning, a subset of machine learning, relies on multi-layered neural networks to analyse and learn from large, complex datasets. Models like Convolutional Neural Networks (CNNs) are suited for spatial and image data, while Recurrent Neural Networks (RNNs) handle sequential information such as text or time series. What sets deep learning apart is its ability to detect intricate patterns in raw data with little manual feature engineering. This capability makes it particularly effective in mental health prediction, where input data may include written responses, social media activity, or information from wearable devices [28],[29],[30]. Recent studies have applied deep learning to predict mental health issues among students with promising results. For example, CNNs have been used to analyse student writing for signs of anxiety and depression, while RNNs have tracked stress levels through data from wearable devices. Models like BERT have also proven effective in detecting depressive symptoms in online student communications, such as forum posts and chat messages. In this study, deep learning was implemented in RapidMiner, utilising multiple hidden layers to capture complex, nonlinear patterns within student mental health data.[31],[32].

The basic feedforward equation of a neural network layer is: $a^{(l)} = f(W^{(l)}a^{(l-1)} + b^{(l)}$

where $a^{(l)}$ is the activation at layer l, $W^{(l)}$ is the weight matrix, $b^{(l)}$ is the bias vector, and f is the activation function (e.g., ReLU, sigmoid).

III. Methodology

A. Research Framework

This research presents a structured approach for predicting mental health risk in university students using machine learning. The methodology starts with survey-based data collecting, followed by data preprocessing step such as encoding, cleaning, and balancing. A binary class is used to categorise students as at risk (labelled "0") or not at risk (labelled "1"). The three models are well-trained and built to predict mental health risk on the dataset. The performance of each classification is then evaluated and compared using the accuracy, precision, recall, and F1-score ratios. In parallel, a correlation matrix analysis is performed to determine the positive, negative, and independent correlations between the dataset's features. Additionally, the students identified as at risk (labelled "0") are isolated into a separate dataset and evaluated using K-Means clustering to identify shared subgroup characteristics. These insights aim to support the design of more targeted prevention and support strategies for vulnerable student populations. The complete conceptual workflow is illustrated in Fig. 1, summarising the end-to-end process from raw data acquisition to prediction, analysis, and interpretation.

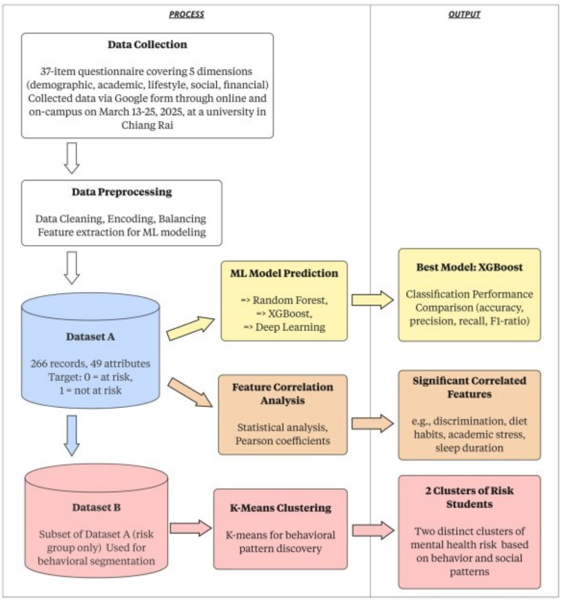

Fig.1. Research Framework

B. Data Collection

The objective of the data collection process was to obtain real-world behavioural and lifestyle information from university students for the purpose of training supervised machine learning models to predict mental health status. The strategy focused on gathering behavioural, educational, and lifestyle data rather than clinical or diagnostic information such as PHQ-9 or GAD-7, to reduce stigma, increase participation, and ensure broader accessibility among students. A structured Google Form with 37 questions divided into five thematic categories—demographics, academic performance, lifestyle choices, social interaction patterns, and financial considerations—was used to gather the data. These segments were developed based on an extensive literature review of previous studies in student mental health prediction to ensure relevance and coverage of influential variables [10],[33]. A hybrid data collection approach was used to ensure a representative diverse dataset. Two-thirds of the responses were obtained on campus through in-person interactions in public spaces, with the remaining one-third coming from online distribution such as social media and university groups. This dual approach improved the overall response rate and mitigated sample bias that could occur if only digitally active students were surveyed. Data was collected at a university in Chiang Rai during a period of one week, from March 13 to March 20, 2025. Participants came from a variety of academic programs and ranged in age from 18 to 25. The survey was designed to be brief (requiring no more than 15 minutes to complete), anonymous, and non-invasive. At the start of the survey, there was a consent statement that included information about the study's goal, voluntary participation, and confidentiality protection. Instead of using direct clinical tests like the PHQ-9 or GAD-7, the questions focused on behaviours that could be observed and measured, like screen time, exercise frequency, and sleep duration [22],[34]. To further improve the dataset's inclusivity and richness, questions about stress coping mechanisms and encounters with discrimination (such as based on gender identity or physical appearance) were added. The self-reported mental health status in one question of the survey, which performed as the target variable for classification tasks, is a crucial component of the dataset. This self-report survey method had drawbacks even though it allowed for broad participation and scalability. Due to self-report bias, students may overestimate or underestimate their actions or emotional states, which might affect the accuracy of the answers. Furthermore, no clinical verification or behavioural tracking was done. Therefore, rather than being used for diagnostics, the dataset is best suited for training models meant for early detection or pre-screening.

C. Data Preparation

The dataset used in this study consisted of 266 responses collected through a structured survey of university students. Following the removal of non-essential metadata such as timestamps, a final dataset of thirty-three attributes, including behavioural, academic, social, lifestyle, and financial data, was retained. The self-reported mental health status is set as a labelled attribute of the dataset where its options are ranged from 1 (poorest) to 5 (excellent). This was transformed into a binary label: values 1, 2, and 3 were grouped as 'at risk', and 4 and 5 as 'no risk'. The resulting class distribution was slightly imbalanced, with 147 'at risk' and 119 'no risk' responses. In this case, the SMOTE technique is used to increase model generalisability and training stability. SMOTE was used before model training. The k-nearest neighbour was performed under SMOTE up-sampling technique; therefore, the value of k is set to 2 with normalised ranges transformation to binary, and the up-sampling size is set to 500.

D. Correlation Matrix Analysis

To determine the associations among the key factors for a student's mental health, we conducted correlation analysis among behavioural, psychological, and lifestyle factors. The goal was to determine which factors are most strongly associated with indicators of mental health, thereby informing further analysis and predictive model development.

E. Custering with K-Means

We utilised the K-means clustering algorithm, an unsupervised machine learning technique that groups comparable data points based on their attributes, to identify underlying behaviours in students' mental health data.

We used K-means clustering on a dataset that only included data entries categorised as poor mental health or at risk (labelled as “0"). By setting k to 2, the algorithm divides data points into two clusters, each with unique behavioural features.

F. Classification

In this stage, the RapidMiner Studio software was used to implement three machine learning classification algorithms—Random Forest, XGBoost, and Deep Learning—for predicting university students' mental health status. Due to the class imbalance in the dataset, the SMOTE upsampling was utilised to generate synthetic samples for the minority class, thereby enhancing class balance and improving model performance. Each classification model was tested using 10-fold cross-validation after being trained on 70% of the dataset for stability and variance reduction; the remaining 30% of the data was used for final performance testing. The systematically parameter fine-tuning for each algorithm was carried out with the help of the optimise Parameters (Grid) operator in RapidMiner. Number of trees, splitting criterion, maximum depth, voting technique, and feature selection method were among the parameters that were optimised for Random Forest. The parameters that were the focus of XGBoost optimisation were learning rate, updater, tree depth, booster type, and other performance indicators. The deep learning model was optimised using learning rate, standardisation, runtime, and random seed parameters. Several parameters were varied during optimisation, while others were kept constant to evaluate their effect on model performance. All three models were assessed for their classification performance based on accuracy, weighted mean recall, and weighted mean precision.

G. Performance Comparasion

This study used four standard metrics generated from the confusion matrix—accuracy, precision, recall, and F1-ratio—to assess and compare the classification performance of the models. Prediction results are broken down into four categories by the confusion matrix: True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN), which represent cases that were correctly or incorrectly identified. Specifically, FP is a Type I error, indicating that a non-affected student is mistakenly identified as at risk, but FN is a Type II error, indicating that a student who is truly in danger is not detected. The evaluation metrics are as follows: recall indicates the ability to capture true positives; precision indicates the ability to prevent false positives; accuracy measures the percentage of overall correct predictions; and the F1-score balances precision and recall, highlighting trade-offs between sensitivity and specificity.

$$Accuracy = ((TP+TN))/((TP+FP+FN+TN)) \quad (1)$$

$$Precision = TP/((TP+FP)) \quad (2)$$

$$Recall = TP/((TP+FN)) \quad (3)$$

$$F1 = ((2*Precision*Recall))/((Precision + Recall)) \quad (4)$$

IV. RESULTS

A. Statistical Data Analysis

The dataset's descriptive analysis discovered several tendencies related to the mental health issues aligned with the biopsychosocial framework which emphasises the interaction of behavioural, emotional, and environmental factors [35]. Respondents who were classified as at risk reported higher academic workloads, frequently surpassing 30 hours per week, and lower average sleep durations (mean = 5.8 hours) than those who were not at risk (mean = 6.2 hours). This raises the possibility of a connection between poor mental health, academic pressure, and insufficient sleeping. In terms of coping strategies, over 65% of students indicated applying passive methods such as watching TV, playing video games, or browsing social media. Passive copings were more common among at-risk students, while only about 30% of the students used proactive coping strategies like creative hobbies or exercise. These behaviours correlate with diminished resilience and align with cognitive-behavioural theories highlighting the role of coping mechanisms in mental distress [36]. Furthermore, lower confidence in managing workloads and academic support (mean scores below 3.2) among at-risk students suggests reduced perceived self-efficacy, a psychological factor often linked to anxiety and academic burnout. Substance use (caffeine, medications) and social-emotional factors such as loneliness and negative social comparison also showed higher prevalence and moderate correlation with mental health risk (r = +0.25 to +0.35), reinforcing the influence of external stressors in the sociocultural dimension. These results validate the relevance of chosen factors for machine learning-based early risk detection and experimentally verify the theoretical argument that multidimensional stressors—academic, behavioural, and psychosocial—converge to impact students' mental health.

B. *Correlation Results*

TABLE I. CORRELATION ANALYSIS RESULTS

Factor 1	Factor 2	Correlation Coefficient (r)
Experience with Discrimination Based on Physical Appearance at University	Experience with Discrimination Based on Gender Identity or Sexual Orientation	0.512
Work While studying	Finance Part-time Job	0.509
Confidence in managing academic workload	Stress management through watching TV, playing games, or scrolling social media	0.00
Managing stress by ignoring or avoiding the problem	The use of antidepressants, whether prescribed or self-medicated	0.00
Daily Diet Mix Food	Daily Diet Fast Food	-0.718
Finance Family Support	Finance Part-time Job	-0.606

From the results of the correlation analysis, we have identified six significant correlations, which are two positive, two negative correlation, and two independent factors respectively, as shown in table I. A moderate positive correlation (r = 0.512) implies that students who have faced physical appearance discrimination are more likely to face prejudice based on gender identity or sexual orientation at university. Moreover, a strongly positive correlation (r = 0.509) suggests that students who work while studying are more likely to be employed in part-time occupations. A significant negative correlation (r = -0.718) was found between students who follow a diverse diet and those who eat fast food daily. Similarly, students with good parental financial support are less likely to work part-time, as seen by the connection (r = -0.606). Furthermore, confidence in managing academic workload showed no correlation with stress management through watching TV, playing games, or scrolling social media (r = 0.000). Similarly, managing stress by ignoring or avoiding the problem showed no correlation with the use of antidepressants, whether prescribed or self-medicated (r = 0.000). These findings indicate that these coping strategies are not statistically associated with the respective variables.

C. *Clustering Results*

The K-means algorithm divided students into two significant clusters, each showcasing a distinct behavioural style despite highlighting poor mental health scores. As Fig. 2, the cluster one occupies students with poor mental health but demonstrates higher social engagement, healthier habits, and adaptive coping tactics. The group can be named "Struggling but Socially Coping." In the second cluster, students with poor mental health can be found and are characterised by higher academic pressure, greater isolation, and unhealthy lifestyle choices. These students can be labelled as "highly stressed and isolated." Despite being in the same severity range, these clusters provide an in-depth comprehension of the diverse ways in which mental health challenges become apparent among students.

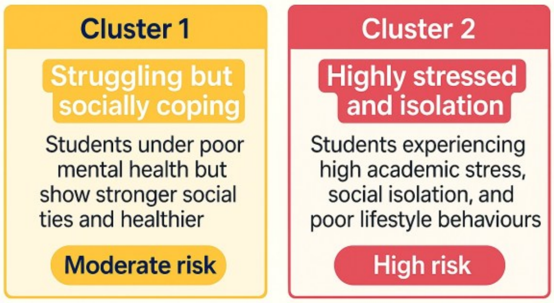

Fig.2. Custers of Mental Health at Risk Students.

D. *Classification Performance Results*

To find out how well the models classified mental health disorders, they were evaluated using accuracy, precision, recall, and F1-score. XGBoost outperformed the other models in terms of overall performance. It obtained an accuracy of 90.60% with a precision of 90.44%, a recall of 78.35%, and an F1 score of 83.90%. This shows that XGBoost is the most reliable classifier on our dataset in this investigation since it not only performs exceptionally well in accurately detecting positive situations but also keeps a comparatively low false negative rate.

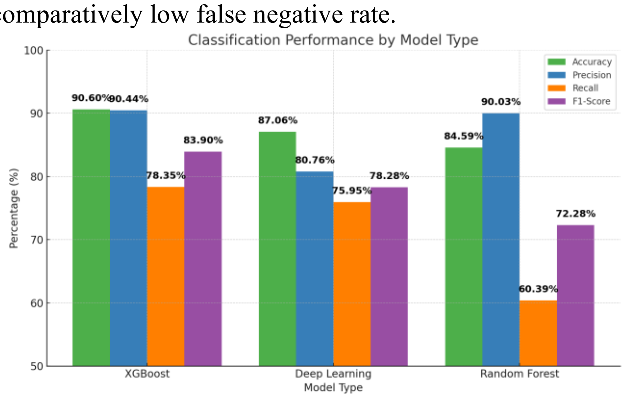

Fig.3. Classification Performance Result.

The deep learning model's accuracy, precision, and recall were 87.06%, 80.76%, and 75.95%, respectively. Despite being marginally less predictive overall than XGBoost, these measures produced an F1 score around 78.28%, indicating a well-balanced performance between sensitivity and specificity. Although the Random Forest model achieved a high precision of 90.03% and a reasonable accuracy of 84.59%, its recall was somewhat low at 60.39%, resulting in an F1 score of 72.28%. This implies that while Random Forest was successful in reducing false positives, its recall performance was compromised since it was less successful in detecting all pertinent positive cases. Overall, XGBoost performed better than the other classifiers in this task, showing superiority in the majority of evaluation measures. Random Forest's high precision was limited by its comparatively low recall, while the deep learning model followed closely behind, providing a competitive balance between precision and recall.

Fig. 3 shows the classification performance measures of all classifiers and Table III shows the parameters that were optimised via hyperparameter tuning for the three classifiers.

TABLE II. PARAMETER FINE TUNNING FOR CLASSIFIERS

Random Forest
Number of trees:200, Maximal depth:15, Voting Strategy:confidence vote, Apply random split, apply subset ratio: Yes
XGBoost
Booster type:DART, Rounds:25, Learning rate, min child weight, subsample, lambda:1.0, Tree depth:100, Normalise type:Tree, Early stopping:No, Min split loss, alpha, rate drop, skip drop:0.0
Deep Learning
Activation:Rectifier, Missing values handling:Mean Imputation, Max runtime seconds:100, Max w2, epochs:10.0, train samples per iteration:-2, epsilon:1.0E-8, rho:0.99, L1:1.0E-5, L2:0.0

V. CONCLUSION

The study used ML models like random forest, XGBoost, and deep learning to predict university students' mental health based on academic, lifestyle, social, and financial data. It highlights early detection possibilities through non-clinical features. Key risk factors identified include academic pressure, sleep deprivation, discrimination, and passive coping. Correlation and clustering analyses identified the behaviours and characteristics of at-risk students, suggesting potential for targeted interventions. The study's limitations include a small sample size, self-reporting bias, lack of clinical validation, and reliance on cross-sectional data, which prevents tracking mental health changes over time. Future studies should increase data size, include wearable device data, and refine clustering to better identify mental health subgroups and improve intervention strategies.

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